MIL-C-17/33A
9 October 1979
SUPERSEDING
MIL-C-17/33
7 September 1955

MILITARY SPECIFICATION SHEET

CABLES, RADIO FREQUENCY, FLEXIBLE, COAXIAL, PULSE, 48 OHMS, M17/33-RG64

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the cables described herein shall consist of this document and the latest issue of Specification MIL-C-17.

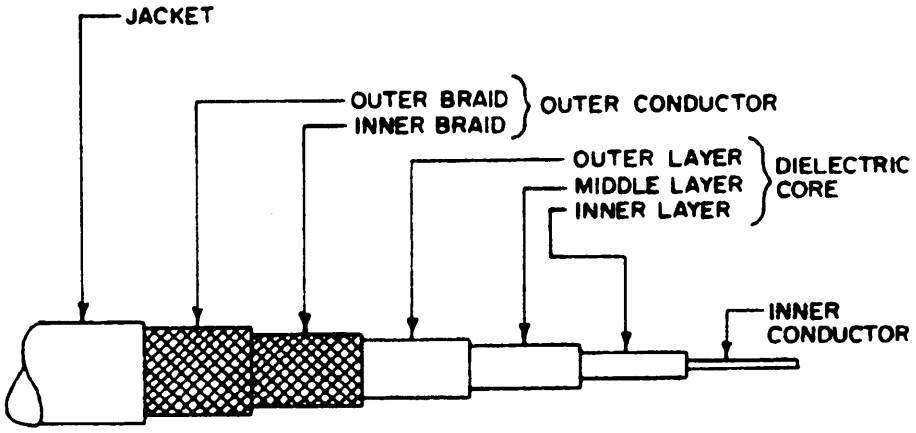


FIGURE 1. Configuration.

FSC 6145

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T16:01Z
Check the source to verify that this is the current version before use.

TABLE I. Description.

Components	Construction details
Inner conductor	Nineteen strands of tinned, copper wire, each strand .0117 inch diameter. Nominal diameter: .0585 inch ±.0020.
Dielectric core Inner layer Middle layer Outer layer	Type E: Composite of three layers. Type H: Semiconducting synthetic rubber. Diameter: .090 inch ±.010. Type W: Insulating rubber. Diameter: .248 inch ±.010. Type Q: Insulating neoprene rubber. Diameter: .288 inch ±.010.
Outer conductor Inner Braid	Double braid of AWG #34, tinned, copper wire. Diameter: .368 inch maximum. Coverage: 97.5%, nominal Carriers: 16 Ends: 10 Picks/inch: 11.7 ±10%
Outer Braid	Coverage: 94.6%, nominal Carriers: 24 Ends: 7 Picks/inch: 14.6 ±10%
Jacket	Type IV Diameter: .460 inch ±.015.

Engineering information:

Continuous working voltage: 7,500 V peak.
Operating frequency: 1 GHz, maximum.
Velocity of propagation: 49.5 percent, nominal.
Operating temperature range: -40° to +85°C.
Inner conductor properties:
DC resistance (maximum at 20°C): .482 ohms per 100 feet.
Elongation: 15% minimum.
Engineering notes: This cable useful in pulse applications (see connector per MIL-C-3607).

REQUIREMENTS:

Dimensions, configuration, and descriptions: See figure 1 and table I.

Environmental and mechanical:
Adhesion of conductors:
Inner conductor to core: 4 pounds minimum, 8 pounds maximum.
Aging stability: +98°C ±2°C.
Stress crack resistance: Not applicable.
Outer conductor integrity: Not applicable.
Dimensional stability: +85°C ±2°C.
Inner conductor from core: .062 inch, maximum.
Inner conductor from jacket: .125 inch, maximum.
Contamination: Not applicable.
Bendability: Not applicable.
Flammability: Not applicable.
Weight: 22 pounds per 100 feet, maximum.

Electrical:
Spark test: 5,800 Vrms +25%, -0%.
Voltage withstanding: 3,000 Vrms.
Insulation resistance: 500 megohms, minimum.
Corona extinction voltage: 10,000 Vrms, minimum.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T16:01Z
Check the source to verify that this is the current version before use.

Characteristic impedance: 48 ohms ±4.
Attenuation: 0.7 dB per 100 feet maximum at 1 MHz.
Structural return loss: Not applicable.
Capacitance: 55 pF per foot, maximum.
Capacitance unbalance: Not applicable.
Transmission unbalance: Not applicable.
Mechanically induced noise voltage: Not applicable.
Time delay: Not applicable.

Part number: See table II.

Supersession data: See table II.

TABLE II. Cross reference of part number.

Part number	Superseded part number or type designation
M17/33-RG64	RG64A/U

NOTE: Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - SH
Air Force - 11, 17, 99
DLA - ES, IS

User activities:
Army - ME, AT, AR
Navy - AS, OS, MC
Air Force - 19

Preparing activity:
Army - CR

Agent:
DLA - ES

(Project 6145-0716-6)

Source: https://assist.dla.mil -- Downloaded: 2026l0-15T16:01Z
Check the source to verify that this is the currentversion before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T16:01Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

HQ, USA CORADCOM
ATTN: DRDCO-CM-DM
Fort Monmouth, NJ 07703

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T16:01Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

Source: https://assist.dla.mil -- Downloaded: 2026l0-15T16:01Z
Check the source to verify that this is the current version before use.